

Procedure for finding Miller Indices

TRWP

- * Find the intercepts of the plane along three axes.
- * Take reciprocal of these intercepts
- * Reduce the reciprocal into whole numbers.
- * Enclose these whole numbers into paranthesis to get Miller Indices
- * The following points are important for getting Miller Indices.

- If the plane is parallel to any axis, its intercept is infinity (∞)
∴ The Miller index is zero (0).

- If a plane cuts negative side of the axis, put a bar (-) over the miller index.

Eg: $[\bar{1}00]$

Miller Indices

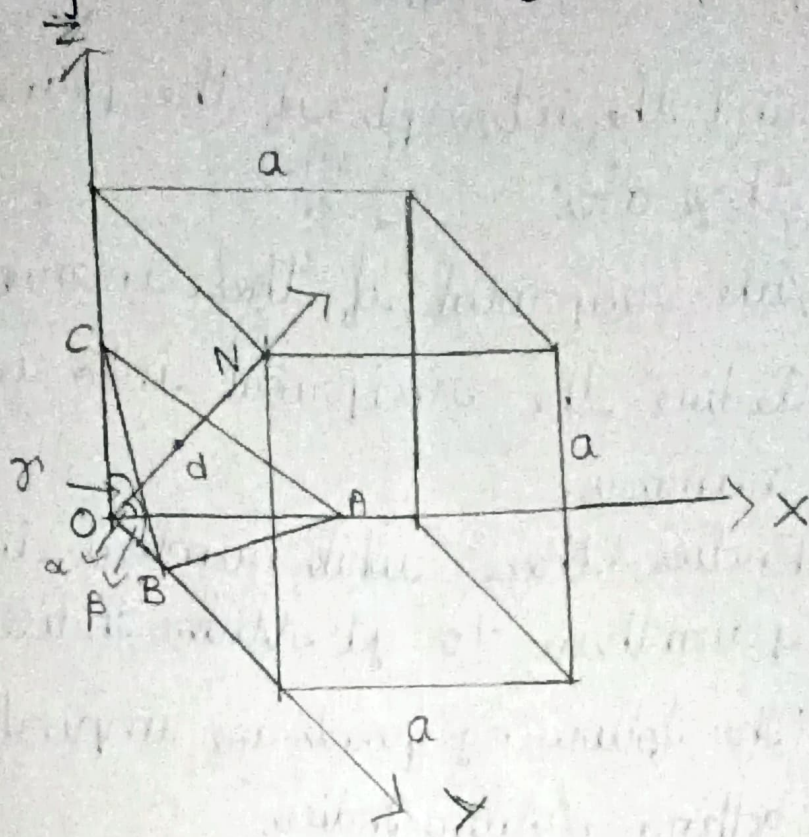
Numbers (hkl) that show the direction and orientation of the plane in a crystal.

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Numbers that show the direction and orientation of the plane in the crystal.

d - Spacing in Cubic Lattice



Consider a cubic crystal. Let ' a ' be the sides of the cube. Consider a plane ABC . Draw a perpendicular line ON from origin to plane ABC . It represents interplanar distance. Let OA , OB , OC be the intercepts on the plane x , y , z and their angles are α , β , γ .

Miller indices of the plane

$$OA : OB : OC = \frac{1}{h} : \frac{1}{k} : \frac{1}{l}$$

Since 'a' is the sides of the cubes,

$$OA : OB : OC = \frac{a}{h} : \frac{a}{k} : \frac{a}{l}$$

$$\therefore OA = \frac{a}{h}$$

$$OB = \frac{a}{k}$$

$$OC = \frac{a}{l}$$

From figure,

$$\cos \alpha = \frac{ON}{OA} = \frac{d}{a/h} = \frac{dh}{a}$$

$$\cos \beta = \frac{ON}{OB} = \frac{d}{a/k} = \frac{dk}{a}$$

$$\cos \gamma = \frac{ON}{OC} = \frac{d}{a/l} = \frac{dl}{a}$$

By the law of direction of cosines,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\left(\frac{dh}{a}\right)^2 + \left(\frac{dk}{a}\right)^2 + \left(\frac{dl}{a}\right)^2 = 1$$

$$\frac{d^2}{a^2} (h^2 + k^2 + l^2) = 1$$

$$d^2 = \frac{a^2}{h^2 + k^2 + l^2}$$

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$